Concoction of *Andrographis paniculata* and eight other herbs (Nilavembu Kudineer) is effectively used to treat malaria and dengue.

Turmeric

Botanical name : *Curcuma longa*

Family: Zingiberaceae

Origin and Area of Cultivation: You have already studied the details under the spices.

Active principle: Curcumin.

Medicinal importance:

Curcumin (the yellow colouring principle is the major pharmacologically active compound of turmeric) is well known for its medicinal properties.

It is used to treat **Alzheimer's** disease due to its property to cross over blood brain barrier. It has a very powerful anti-oxidant, anti-cancerous, anti-inflammatory, anti-diabetic, anti-bacterial, anti-fungal and anti-viral properties. It is one of the traditional medicines used for wound healing.



Patenting Of Turmeric

University of Mississippi medical center, USA was granted a patent for wound healing property of Turmeric in 1995. The patent was granted both for oral and topical applications and provides an exclusive right to sell and distribute. Since the use of turmeric to heal wounds is a common domain knowledge in India, the Government of India has decided to fight against the patent through the Indian Council for Scientific and Industrial Research (CSIR). CSIR collected documentary evidences from various literature to prove that the knowledge on wound healing property of turmeric existed in India for a long time and provided the evidences to the United States Patent and Trade mark Office (USPTO). Based on the evidences the patent was revoked by USPTO. Hence the traditional knowledge (TK) on turmeric was safeguarded from Bio piracy.

Psychoactive Drugs

In the above chapter you have learnt about plants that are used medicinally to treat various diseases. Phytochemicals / drugs from some of the plants alter an individual's perceptions of mind by producing hallucination are known as psychoactive drugs. These drugs are used in all ancient culture especially by Shamans and by traditional healers. Here we focus on two such plants namely Poppy and Marijuana.

Opium poppy

Botanical name : *Papaver somniferum*

Family: Papaveraceae

Origin and Area of cultivation: *Opium poppy* is native to South Eastern Europe and Western Asia. Madhya Pradesh, Rajasthan and Uttar Pradesh are the licenced states to cultivate opium poppy.

Opium is derived from the exudates of fruits of poppy plants. It was traditionally used to induce sleep and for relieving pain. Opium yields **Morphine**, a strong analgesic which is used in surgery. However, opium is an addiction forming drug.

Cannabis / Marijuana

Botanical name : *Cannabis sativa*

Family: Cannabiaceae

Origin and Area of Cultivation: Marijuana is native to China. States such as Gujarat, Himachal Pradesh, Uttarkand, Uttarpradesh and Madhaya Pradesh have legally permitted to cultivate industrial hemp/Marijuana

The active principle in Marijuana is **trans-tetrahydrocanabinal** (THC). It possess a number of medicinal properties. It is an effective pain reliever and reduces hypertension. THC is used in treating **Glaucoma** a condition in which pressure develops in the eyes. THC is also used in

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